Soft Condensed Matter: An Introduction (2025)

Lectures 7-8: Student Presentations of Foundational and Recent Papers

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Web version

Session Format

Mini-Seminars

Structure:

- Each student presents one selected paper
- **10–12 minutes** per presentation
- Followed by peer discussion and Q&A (3 minutes)
- Instructor moderates and connects to course themes

Objectives:

- Expose class to broad range of soft matter topics
- Practice critical reading skills
- Develop scientific presentation abilities
- Engage in scholarly discourse

Presentation Guidelines

What to Include

1. **Paper Context**
    - Authors, journal, year
    - Historical or scientific significance
    - Connection to course themes
2. **Main Questions/Hypotheses**
    - What problem does paper address?
    - What was known before?
    - What gap does it fill?
3. **Key Methods**
    - Experimental techniques OR
    - Theoretical approach OR
    - Computational methods

- Briefly explain novel techniques

4. **Main Results**
   - Most important findings
   - Key figures/data
   - Quantitative results when relevant
5. **Significance**
   - Why does this matter?
   - Impact on field
   - Applications or implications
   - Connection to modern research
6. **Critical Thinking**
   - Assumptions made
   - Limitations
   - Future directions

Presentation Tips

- **Visual:** Use slides with clear figures
- **Clarity:** Explain jargon and technical terms
- **Focus:** Don't try to cover everything
- **Engagement:** Make eye contact, invite questions
- **Time:** Practise to finish within 10–12 minutes

Peer Discussion

Students should:

- Ask clarifying questions
- Pose critical questions
- Make connections to other papers/lectures
- Respectful and constructive

Example Questions:

- "What are assumptions of that model?"
- "How might experiment behave with different fluid?"
- "How does this connect to [course topic]?"

Paper Topics (10 Options)

Paper 1: Brownian Motion and Atomic Theory

Citation: Einstein, A. (1905) "On the motion of small particles suspended in stationary liquids required by the molecular-kinetic theory of heat."

Key Points:

- Explained jittery motion of pollen grains as molecular collisions
- Provided direct evidence of atoms
- Quantified diffusion
- Related diffusion coefficient to particle size and temperature
- Jean Perrin experimentally validated predictions (1908)

Why Foundational:

- Cornerstone of soft matter physics
- Introduced stochastic processes underlying colloidal science
- Bridge between microscopic and macroscopic

Connection to Course:

- Lecture 1: Thermal energy scale, mesoscopic structure
- Underpins all subsequent colloidal physics

Paper 2: Entropic Forces and Liquid Crystal Ordering

Citation: Onsager, L. (1949) "The effects of shape on the interaction of colloidal particles" Ann. N.Y. Acad. Sci.

Key Points:

- Theory of isotropic-nematic transition in rod-like colloids
- Purely repulsive hard rods can spontaneously align at high concentration
- Driven by **entropy maximization**
- Increased orientational order frees up translational entropy
- Drives nematic liquid crystal phase

Why Foundational:

- Generalized how shape alone causes phase transitions
- Laid groundwork for liquid crystal science
- Polymer liquid crystals
- Counter-intuitive: more order increases entropy!

Connection to Course:

- Lecture 1: Entropy-driven behavior in soft matter
- Shows complexity emerging from simple interactions

Paper 3: Polymer Dynamics (Reptation)

Citation: de Gennes, P.G. (1971) "Reptation of a Polymer Chain in the Presence of Fixed Obstacles" J. Chem. Phys.

Key Points:

- Reptation model for polymer melts

- Polymer moves snake-like through tube of confining neighbors
- Explains dramatic viscosity increase in long entangled polymers
- Specific scaling: diffusion ~ N^{-2} (N = polymer length)
- Reptation length ~ N^{1}

Why Foundational:

- Revolutionized polymer rheology
- Molecular picture for viscoelasticity
- Earned de Gennes Nobel Prize (1991) for soft matter universality

Connection to Course:

- Lecture 5: Viscoelastic behavior of polymers
- Lecture 6: Relaxation time and the Deborah number

Paper 4: Auxetic Materials

Citation: Lakes, R.S. (1987) "Foam Structures with a Negative Poisson's Ratio" Science

Key Points:

- Fabricated novel re-entrant foam
- Expands laterally when stretched (Poisson's ratio = –0.6)
- Counter-intuitive behavior from internal structure design
- Hinged cell ribs create auxetic behavior
- Improved stiffness and energy absorption

Why Foundational:

- Opened field of mechanical metamaterials
- Properties thought impossible in natural materials
- Design-driven material properties

Connection to Course:

- Soft matter engineering
- Structure-property relationships
- Mesoscopic design principles

Paper 5: Colloidal Glass Transition

Citation: Pusey, P.N. & van Megen, W. (1986) "Phase Behavior of Concentrated Suspensions of Nearly Hard Colloidal Spheres" Nature

Key Points:

- Demonstrated glass transition in colloidal hard-spheres
- Dramatic slowdown in particle motion at high volume fraction

- Structural arrest using dynamic light scattering
- Approaches random close packing (~58% volume fraction)
- Analogous to molecular glass formation

Why Significant:

- Established colloids as model for glassy dynamics
- Bridge soft matter and condensed matter
- Disorder-induced solidification is general phenomenon
- Colloids allow real-space imaging of glass transition

Connection to Course:

- Soft matter phase transitions
- Role of volume fraction and packing
- Non-equilibrium phenomena

Paper 6: Jamming Concept

Citation: Liu, A.J. & Nagel, S.R. (1998) "Jamming is Not Just Cool Any More" Nature

Key Points:

- Unified disparate systems: granular media, glasses, foams
- Proposed jamming phase diagram (density, load, temperature)
- Above certain density + below certain temperature/shear → jammed (rigid)
- "Fragile matter" rigidifies under stress
- Collapses if stress direction changes

Why Important:

- Interdisciplinary unification
- How liquids solidify without crystalline order
- Connected granular physics with phase transitions
- Spawned new research field

Connection to Course:

- Lecture 5: Solid-like versus liquid-like response
- Transition from fluid to solid
- Shear thickening (related to paper 10)

Paper 7: Active Matter

Citation: Vicsek, T. et al. (1995) "Novel Type of Phase Transition in a System of Self-Driven Particles" Phys. Rev. Lett.

Key Points:

- Simple model of flocking behavior

- Point particles: constant speed + align with neighbors + noise
- Continuous phase transition from disordered to collective motion
- Non-zero average velocity when noise reduced or density increased
- Now called "Vicsek model"

Why Significant:

- Foundational in active matter field
- Local alignment → macroscopic order
- Applies to bird flocks, fish schools, self-propelled colloids
- Spontaneous symmetry breaking in non-equilibrium systems

Connection to Course:

- Non-equilibrium soft matter
- Collective behavior
- Pattern formation
- Beyond thermal equilibrium

Paper 8: Mechanobiology

Citation: Engler, A.J. et al. (2006) "Matrix Elasticity Directs Stem Cell Lineage Specification" Cell

Key Points:

- Stem cells sense substrate stiffness
- Commit to different fates based on mechanical cues
- Soft (~0.1-1 kPa, brain-like) → neurons
- Intermediate (~10 kPa, muscle-like) → muscle cells
- Stiff (~30-40 kPa, bone-like) → osteoblasts
- Inhibiting cytoskeletal tension (myosin II) prevents effect

Why Groundbreaking:

- Linked soft matter mechanics to biology
- Tissue softness directs cell differentiation
- Profound implications for tissue engineering
- Understanding developmental biology

Connection to Course:

- Lectures 5–6: The elastic modulus as a control parameter (kPa range matters!)
- Biological relevance of "softness"
- Cells as mechanical sensors

Paper 9: Soft Robotics

Citation: Shepherd, R.F. et al. (2011) "Multigait soft robot" Proc. Natl. Acad. Sci.

Key Points:

- Completely soft pneumatic robot from Whitesides' group
- Made of silicone elastomers
- Could crawl and walk with different gaits
- Switch between undulating snake-like and four-legged walking
- Controlled by air inflation sequences
- Could sustain impacts and squeeze through gaps

Why Notable:

- Seminal demonstration of soft robotics
- Functionalities hard for rigid robots
- Safe human interaction
- Morphing locomotion
- Inspired wave of soft robotics research

Connection to Course:

- Soft matter engineering
- Rubber elasticity applications
- Pneumatic networks
- Search-and-rescue, medical devices

Paper 10: Shear Thickening & Dynamic Jamming

Citation: Waitukaitis, S.R. & Jaeger, H.M. (2012) "Impact-Activated Solidification of Dense Suspensions via Dynamic Jamming Fronts" Nature

Key Points:

- Investigated cornstarch-water suspension ("oobleck")
- Turns solid-like under impact
- High-speed imaging + force measurements
- Rapidly moving object creates **jamming front** in suspension
- Grains jam into temporary rigid network
- Once front percolates, material solidifies locally

Why Significant:

- Mechanism for discontinuous shear thickening
- Links to transition from unjammed to jammed under stress
- Explains everyday phenomena (running on oobleck!)
- Novel impact-resistant materials
- Shear-thickening fluids in protective gear

Connection to Course:

- Lectures 5–6: Non-Newtonian flow
- Lecture 3: Instabilities and pattern formation
- Dramatic material behavior change under stress

Schedule Overview

The final running order is set after paper selection. Each slot is about 15 minutes: 10–12 minutes to present and about 3 minutes for questions.

Instructor Role

During Presentations

- Time keeping
- Note connections to course material
- Prepare bridging comments

After Each Presentation

Connect to Course Concepts:

- Example: "Einstein (1905) relates to Brownian motion and energy scales from Lecture 1"
- Example: "Engler et al. (2006) ties into the importance of the elastic modulus from Lectures 5–6"

Facilitate Discussion:

- Encourage questions
- Prompt deeper thinking
- Ensure respectful dialogue

Wrap-Up

Course Synthesis:

- Survey broad swath of soft matter
- 100-year-old theories to cutting-edge research
- Versatility and unifying principles of field
- From fundamental physics to engineering applications
- From molecular scale to macroscopic behavior

Learning Outcomes

By end of these sessions, students will:

1. **Breadth of Soft Matter:**
    - Appreciate historical development
    - Understand current research frontiers
    - See connections across sub-disciplines
2. **Critical Reading:**

- Extract key points from scientific papers
- Evaluate assumptions and limitations
- Place work in broader context

3. **Scientific Communication:**
   - Present complex ideas clearly
   - Respond to questions professionally
   - Engage in scholarly discussion
4. **Course Integration:**
   - Connect papers to lecture themes
   - Synthesize knowledge across topics
   - Develop holistic understanding

Assessment Criteria

This final presentation contributes **60% of the course grade**. The percentages below describe how the presentation mark itself is calculated.

Presentation (70%)

- **Content Understanding (30%):** Grasp of paper's main ideas
- **Clarity of Explanation (20%):** Clear, organized presentation
- **Visual Aids (10%):** Effective use of slides/figures
- **Time Management (10%):** Finish within 10–12 minutes

Discussion Participation (30%)

- **Questions Asked (15%):** Thoughtful questions to peers
- **Answers Given (15%):** Thoughtful responses to questions

References

Full citations are provided in the Student Presentation Topics document.

Previous Lecture: Lecture 6: Sheets, Bubbles and Drops

Course Home: Course Overview

Paper Selection: Detailed Paper Topics

Metadata

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